



encoding

D3.2 Report on Pattern Identification of Critical Combustion Events

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Contents



1 Introduction

Understanding critical combustion events is a complex task that involves extensive analysis of both experimental and numerical data. Finding patterns in these datasets is an important step towards building better models for combustion phenomena. Many tools exist both in the reactive and non reactive flows. Proper Orthogonal Decomposition (POD) (CITAZIONE), is an Singular Value Decomposition (SVD) based method that provide a low rank representation of the flow rewriting it as a sum of orthogonal spatial patterns. Principal component analysis has been successfully employed to project high-dimensional data onto a lower-dimensional orthogonal subspace of thermodynamical variables (CITAZIONE), effectively reducing the complexity of the system while retaining its essential features. However, this method is not able to capture the temporal evolution of the system and it is not able to provide a physical interpretation of the modes. Dynamic Mode Decomposition (DMD) (CITAZIONE) allows the reduction of the system in spatio-temporal modes which have both a spatial structure and an associated temporal frequency and growth rate. Thus the system is represented as the sum of modes evolving with a linear dynamics in time that allows the study of the spectral properties of the system. Although powerful this method suffers from noisy data and it often is inadequate to provide an accurate description of turbulent reactive flows. High Order Dynamic Mode Decomposition (HODMD) (CITAZIONE) is a extension of DMD that improves the method by introducing the higher order Koopman hypothesis. This makes it more robust and has already been successfully applied in reactive flows (CITAZIONE) Combustion data often presents itself in tensorial form. Moreover the most valuable datasets, i.e. the ones that most closely represent the real physics of the system, are often the ones with the highest dimensionality both in spatial component and in the number of variables. This has sparked interest in methods that process data directly into this form and don't require reshaping into matrix form as in the case of SVD based methods and DMD.

The current report focuses on describing the application of these techniques to enhance the algorithms that allows pattern identification in combustion data. High Order Singular Value Decomposition (CITAZIONE), a generalization of SVD, is discussed as a tool to both compress and extract physical patterns from huge combustion datasets. In particular, this method allows for multi modal analysis along each axis of the tensor at the same time, allowing to quantify the importance of each dimension of the tensor. When combined with HODMD this, allows for identification of key feature in the dynamics of the reactive flow.

Furthermore,

2 Datasets

The methodologies are assessed on different datasets from the consortium. These data describe high fidelity simulations of complex reacting flows.



2.1 NH₃/H₂ non-premixed mixing layer

This dataset consists of direct numerical simulations (DNS) of a two-dimensional non-premixed ammonia-hydrogen-air reacting mixing layer. It is generated by DCGIOVANNI at CORIA using the structured mesh reactive flow solver SiTCom-B [?]. The configuration represents a splitter-plate stabilized flame in which a cold fuel stream and a hot air stream interact across a shear layer, representative of conditions encountered in practical burners.

The lower stream consists of a NH₃/H₂ fuel blend (with 15% H₂ by volume) at ambient temperature (300 K) with a mean velocity of 15 m s⁻¹, while the upper stream consists of preheated air with a mean velocity of 39.6 m s⁻¹. A cold wall maintained at 600 K is imposed at the top boundary, and a symmetry condition is applied at the bottom. Homogeneous velocity fluctuations of 10% are prescribed at both inlets. The computational domain measures 12 × 1 cm and is discretized on a uniform grid of 2400 × 200 cells with a resolution of 50 μm.

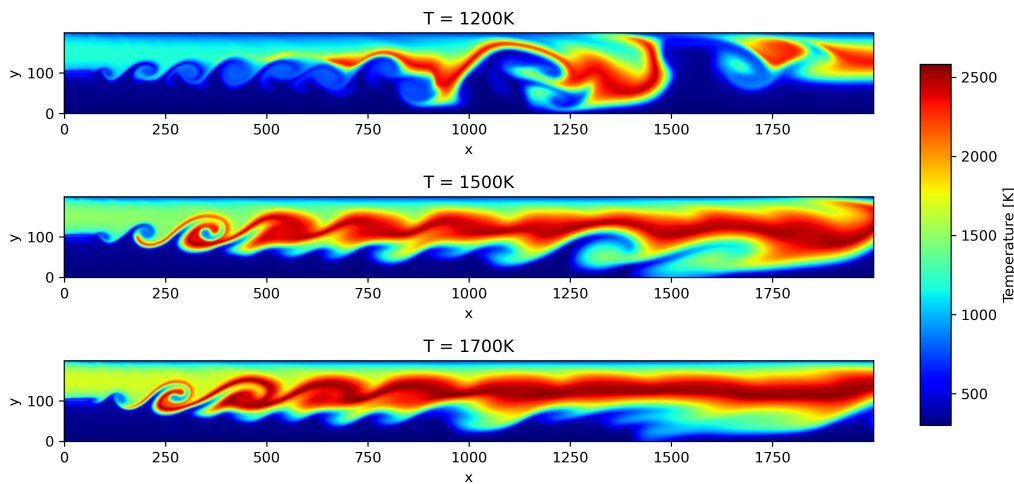


Figure 1: Three representative snapshots of the temperature field from the NH₃/H₂ non-premixed reacting mixing layer DNS, illustrating the spatial evolution of the flame and mixing structures at different instants.

Three datasets are generated at different air preheat temperatures, namely 1200 K, 1500 K, and 1700 K, to probe the sensitivity of the flame dynamics and species evolution to the inlet thermal boundary condition. The domain geometry, grid resolution, and flow velocities are identical across all three cases. Each snapshot contains 19 reactive scalar fields corresponding to the species of the reduced mechanisms RmAH-1 and RmAH-2 [?] — namely H₂, O₂, H₂O, H, O, OH, HO₂, NH₃, NH₂, NH, N, NO, NO₂, N₂O, HNO, NNH, N₂H₂, N₂, and Ar — together with temperature, yielding 20 variables in total. The species analysis presented in this work is conducted exclusively on the 1500 K dataset, which constitutes the reference operating condition for the assessment of the proposed methodologies.

3 Methods

3.1 High Order Dynamic Mode Decomposition

Dynamic Mode Decomposition (DMD) [?] is a data-driven spectral method that extracts spatially coherent structures and their associated temporal dynamics directly from time-resolved snapshot¹ data. Given a sequence of snapshots $\mathbf{v}_k \in \mathbb{R}^J$, $k = 1, \dots, K$, DMD seeks a linear operator $\mathbf{A}$ such that $\mathbf{v}_{k+1} \approx \mathbf{A} \mathbf{v}_k$, whose leading eigenpairs encode the dominant frequencies and growth/decay rates of the system. The full data field is then expressed as a superposition of DMD spatial modes $u_p(\mathbf{x})$ with purely exponential temporal evolution,

$$v(\mathbf{x}, t) = \sum_{p=1}^P a_p u_p(\mathbf{x}) e^{(\delta_p + i\omega_p)(t-1)\Delta t}, \quad (1)$$

where $a_p \geq 0$ are real amplitudes, ω_p are angular frequencies, δ_p are growth rates, and Δt is the sampling interval. Each mode is normalised so that its root-mean-square norm equals unity.

Higher-Order Dynamic Mode Decomposition (HODMD) [?] extends DMD making it more suitable for noisy data and nonlinear phenomena such as the ones encountered dealing with combustion data. The main idea is to extend the assumption that the system's evolution can be described with a linear operator to temporal windows of size d . Each snapshot is thus related to the d subsequent snapshots through a set of linear operators. This generalised assumption leads to a sliding-window construction of an augmented snapshot matrix and allows for better identification of close frequency even in noisy data. When $d = 1$, HODMD reduces exactly to standard DMD.

The HODMD algorithm proceeds in two main steps:

Step 1 — Dimensionality reduction via SVD. A truncated Singular Value Decomposition (SVD) is applied to the original snapshot matrix, compressing the spatial dimension from J to N retained singular values. The tolerance ε_{SVD} controls the level of compression by enforcing

$$\frac{\sigma_j}{\sigma_1} \geq \varepsilon_{\text{SVD}}, \quad (2)$$

where σ_j are the singular values in decreasing order. The resulting temporal coefficients weighted by the retained singular values form the reduced snapshot matrix, which filters noise.

Step 2 — High-order Koopman eigenvalue problem. HODMD is applied to the reduced snapshot matrix with delay parameter d . The eigenvalue problem is solved to extract DMD modes, frequencies and amplitudes. A second tolerance ε_{DMD} selects physically significant modes by retaining only those whose relative amplitude satisfies

$$\frac{a_p}{\max_p(a_p)} > \varepsilon_{\text{DMD}}. \quad (3)$$

¹Matrix that has as columns (rows) the time index and all the other unfolded indices as rows (columns)



3.1 Physical interpretation

From a practical point of view DMD looks for coherent structures that behave in an oscillatory manner across the whole dataset. This makes it easier to observe phenomena that cannot be observed when looking at the original data, that often is very complex. A big issue is that the assumption that the system can be described with a matrix even in snapshot form is very strong and it will fail in case of noisy or turbulent data. HODMD is able to obtain better results because instead of looking for oscillatory coherent structures across the whole dataset, it looks for structures that depend on the last d snapshots instead of just the last one. This makes it possible to extract up to $d \times N$ modes instead of N :

$$\underbrace{\mathbf{v}_{k+1} = \mathbf{A} \mathbf{v}_k}_{\text{DMD}} \longrightarrow \underbrace{\mathbf{v}_{k+d} = \mathbf{A}_1 \mathbf{v}_{k+d-1} + \mathbf{A}_2 \mathbf{v}_{k+d-2} + \dots + \mathbf{A}_d \mathbf{v}_k}_{\text{HODMD}}$$

3.2 Multidimensional Iterative HODMD (mdHODMDit)

3.2 High Order Singular Value Decomposition

Both DMD and HODMD perform truncated SVD for denoising and dimensionality reduction. This effectively filters out structures that have very low temporal contribution across the whole dataset, which often coincides with noise. Combustion data often carry several physical variables and span a multi-dimensional spatial domain. The application of SVD on the reshaped tensor introduces an implicit coupling among modes that belong to different dimensions, and the SVD truncation cannot distinguish noise contributions arising from the spatial, temporal, or variable axes independently. For this reason, a first improvement over the classical version of HODMD is to introduce High Order Singular Value Decomposition (HOSVD) [?] in the first step. This is a generalization of SVD that allows filtering along each dimension of the tensor.

It works on data keeping it in tensor form $\mathcal{T} \in \mathbb{R}^{N_{\text{var}} \times N_x \times N_y \times N_t}$. First, it unfolds $\mathcal{T}$ along each of its four modes in turn, computes the n -mode singular values $\{\sigma_j^{(n)}\}$ from the corresponding unfolded matrix, and retains factor matrices $\mathbf{U}^{(n)}$ by enforcing the truncation criterion

$$\frac{\sigma_j^{(n)}}{\sigma_1^{(n)}} \geq \varepsilon_{\text{tol}}, \quad (4)$$

independently for each mode n . The result is a compressed core tensor $\mathcal{S}$ and a set of factor matrices $\{\mathbf{U}^{(n)}\}_{n=1}^4$ such that $\mathcal{T} \approx \mathcal{S} \times_1 \mathbf{U}^{(1)} \times_2 \mathbf{U}^{(2)} \times_3 \mathbf{U}^{(3)} \times_4 \mathbf{U}^{(4)}$, where $\times_n$ denotes the n -mode product. These matrices are exactly the ones one would obtain performing truncated SVD on each of the unfoldings. Therefore, the temporal factor matrix $\mathbf{U}^{(4)}$ is the same as before.

After the DMD eigendecomposition, the reduced spatial DMD modes are projected back to the original tensor dimensions by contracting with the spatial factor matrices



$\mathbf{U}^{(1)}, \mathbf{U}^{(2)}, \mathbf{U}^{(3)}$ and the core tensor, recovering physically interpretable spatial structures in the full variable- x - y space without ever vectorising the multi-variable field, from this the algorithm is called multidimensional.

Then, the procedure is iterated. The DMD reconstruction is used to form an updated snapshot tensor, which is re-compressed via HOSVD. The spatial ranks r_x, r_y , and r_{var} retained after truncation are monitored: if they differ from those of the previous iteration, the HODMD step is repeated on the new compressed temporal matrix $\hat{\mathbf{V}}_1^K$. Convergence is declared when the spatial ranks no longer change between two successive iterations. This can be interpreted as the low-rank spatial structure implied by the DMD modes is the same as the one identified by HOSVD. The result is a decomposition in which the noise filtering and the dynamic mode identification are consistent, rather than performed independently.

4 Results and Discussion

4.1 mdHODMDit Applied to the CORIA Dataset

This section presents the application of the multidimensional iterative Higher-Order Dynamic Mode Decomposition (mdHODMDit) algorithm to the three NH_3/H_2 DNS datasets from CORIA described in Section ??.

4.1 Data Structure and Preprocessing

The DNS output for each temperature case is stored as a four-dimensional tensor $\mathcal{T} \in \mathbb{R}^{N_{\text{var}} \times N_x \times N_y \times N_t}$, where the axes correspond to the reactive scalar variables, the streamwise spatial coordinate x , the cross-stream coordinate y , and time t , respectively.

Before analysis, the tensor is restricted to the region of physical interest. In the streamwise direction, indices $x_{60:1000}$ are retained, corresponding to the range $[0.3, 5.0]$ cm downstream of the splitter plate, which encompasses the near-field flame stabilisation region and the initial development of the shear layer. In time, the first 74 snapshots are discarded to remove the initial transient, so the analysis is performed on the statistically developed portion of the signal. The temporal resolution is $\Delta t = 40 \mu\text{s}$. For this kind of data scaling is extremely important due to the differences in order of magnitude of the variables considered and the difference between units of measurement [?], [?]. There is no way to determine quantitatively which scaling criteria is best, thus the method was applied to both data scaled with auto scaling and range scaling. The analysis yielded similar results for both scaling methods, identifying the same main frequencies.

For this kind of data, scaling is extremely important due to the differences in the order of magnitude of the variables considered and the differences between their units of measurement [?, ?]. Since there is no universally accepted quantitative criterion to determine which scaling method is most appropriate, the analysis was performed using both auto scaling $x_i^{\text{scaled}} = \frac{x_i - \mu_x}{\sigma_x}$ and range scaling $x_i^{\text{scaled}} = \frac{x_i - x_{\min}}{x_{\max} - x_{\min}}$.



4.1 Parameter Calibration

The analysis is performed over a grid of Hankel delay parameters

$$d \in \{10, 15, 20, 25, 30, 35, 40, 45\} \quad (5)$$

and tolerance values

$$\varepsilon_{\text{tol}} \in \{5 \times 10^{-3}, 2 \times 10^{-2}, 3 \times 10^{-2}, 4 \times 10^{-2}, 5 \times 10^{-2}, 7 \times 10^{-2}, 1 \times 10^{-1}, 2 \times 10^{-1}\}. \quad (6)$$

The same tolerance ε_{tol} governs both the HOSVD truncation (equation (??)) and the amplitude threshold applied to retain DMD modes, so that a single parameter controls the compression level throughout the algorithm.

4.1 Identification of Physical Frequencies

A DMD mode frequency is considered *physical* if the corresponding spectral peak appears consistently across multiple values of the delay parameter d and is not sensitive to the choice of ε_{tol} . Formally, a Strouhal number St_k is flagged as physical if it is identified as a dominant peak (i.e. its normalised amplitude exceeds a fixed threshold) in at least half of the parameter combinations $\{(d_i, \varepsilon_j)\}$ included in the sweep. This robustness criterion filters out artefacts of the decomposition that arise for particular choices of the delay window or compression level, retaining only coherent structures whose temporal periodicity is an intrinsic feature of the flow.

The Strouhal number is defined throughout as

$$St = f \cdot \frac{\ell}{U_b}, \quad (7)$$

where $\ell = 3$ mm is a reference length scale of the mixing layer and U_b is the bulk convective velocity, taken as $U_b = 21.34, 24.29$, and 27.14 m s⁻¹ for the 1200 K, 1500 K, and 1700 K cases, respectively.

4.1 Frequency–Amplitude Spectra

The frequency–amplitude plots shown in Figures ??–?? display, for each temperature case, the normalised DMD mode amplitudes $|\hat{a}_k| / \max_k |\hat{a}_k|$ as a function of Strouhal number St . Each marker corresponds to one $(d, \varepsilon_{\text{tol}})$ pair; different colours encode different values of d and different marker styles encode different tolerances. Peaks that persist across all curves can therefore be read off directly as physically robust frequencies.

Case 1500 K



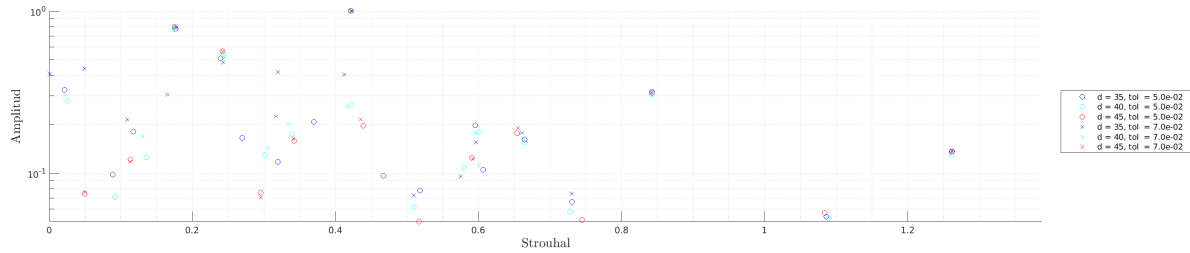


Figure 2: Normalised DMD mode amplitudes as a function of Strouhal number for the 1500 K dataset, for $d \in \{35, 40, 45\}$ and $\varepsilon_{\text{tol}} \in \{5 \times 10^{-2}, 7 \times 10^{-2}\}$. Circles and crosses distinguish the two tolerance levels; colours distinguish the delay values d . Vertical dashes would indicate physically robust peaks identified by the robustness criterion.

Figure ?? reveals several dominant peaks in the Strouhal range $St \approx 0.2$ – 0.5 . The most prominent structures are located near $St \approx 0.21$ – 0.25 , $St \approx 0.27$, $St \approx 0.37$, and $St \approx 0.42$. These peaks are reproducible across all parameter combinations shown, confirming their physical origin. The mode at $St \approx 0.42$ carries the largest amplitude and is therefore identified as the dominant dynamical structure in this configuration.

[PLACEHOLDER — frequency–amplitude spectrum for the 1200 K case]

Figure 3: Normalised DMD mode amplitudes as a function of Strouhal number for the 1200 K dataset, for $d \in \{35, 40, 45\}$ and $\varepsilon_{\text{tol}} \in \{5 \times 10^{-2}, 7 \times 10^{-2}\}$.

Case 1200 K [PLACEHOLDER — describe the dominant Strouhal peaks identified for the 1200 K case, their consistency across parameter combinations, and any qualitative differences with respect to the 1500 K reference.]

Case 1700 K

[PLACEHOLDER — frequency–amplitude spectrum for the 1700 K case]

Figure 4: Normalised DMD mode amplitudes as a function of Strouhal number for the 1700 K dataset, for $d \in \{35, 40, 45\}$ and $\varepsilon_{\text{tol}} \in \{5 \times 10^{-2}, 7 \times 10^{-2}\}$.

[PLACEHOLDER — describe the dominant Strouhal peaks identified for the 1700 K case and qualitative differences with respect to the lower-temperature cases.]

4.1 Spatial DMD Modes (1500 K Reference Case)

The spatial structure of the dominant modes is examined in detail for the reference 1500 K case at $d = 40$, $\varepsilon_{\text{tol}} = 7 \times 10^{-2}$. Each mode is represented as a two-dimensional field over the normalised streamwise–cross-stream plane $(x/h, y/h)$, where $h = 5$ cm is the reference height, for each of the retained reactive scalar fields. Figures ??–?? show the real part of the five dominant modes.

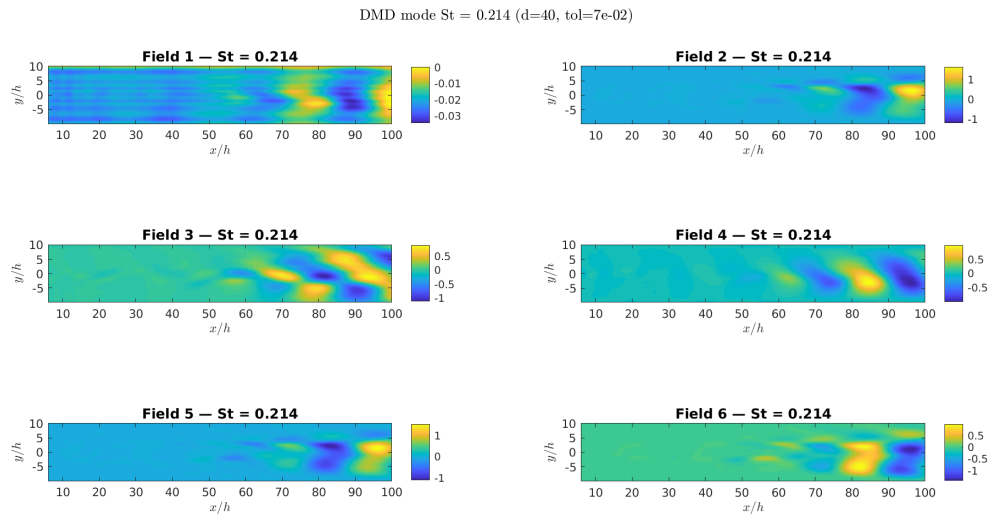


Figure 5: Spatial DMD mode at $St = 0.214$ ($d = 40$, $\varepsilon_{\text{tol}} = 7 \times 10^{-2}$, 1500 K case). Each panel shows the real part of the mode for one reactive scalar field over the streamwise–cross-stream plane.

The mode at $St = 0.214$ (Figure ??) is the lowest-frequency dominant structure identified in the spectrum. It exhibits elongated streamwise coherence with moderate cross-stream extent, consistent with a large-scale shear-layer oscillation associated with the

convective growth of the Kelvin–Helmholtz instability of the mixing layer.

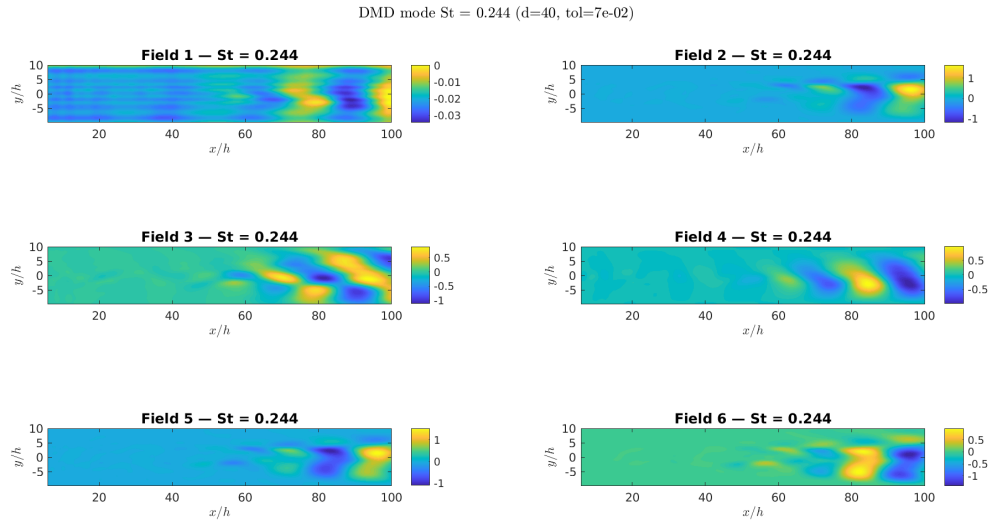


Figure 6: Spatial DMD mode at $St = 0.244$ ($d = 40$, $\varepsilon_{\text{tol}} = 7 \times 10^{-2}$, 1500 K case).

The mode at $St = 0.244$ (Figure ??) presents a spatial organisation similar to the $St = 0.214$ mode but with a shorter streamwise wavelength, suggesting a harmonic or sub-harmonic relationship possibly linked to vortex pairing in the developing shear layer.

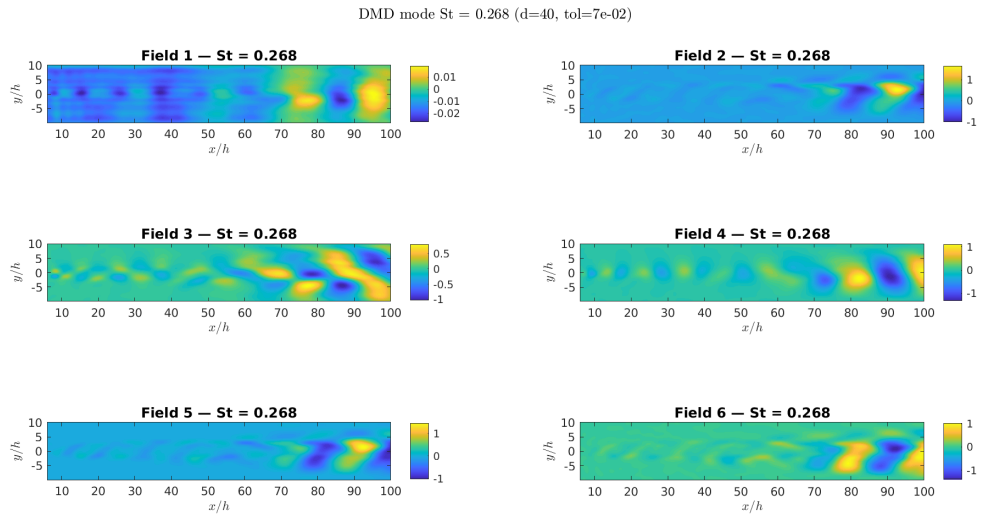


Figure 7: Spatial DMD mode at $St = 0.271$ ($d = 40$, $\varepsilon_{\text{tol}} = 7 \times 10^{-2}$, 1500 K case).

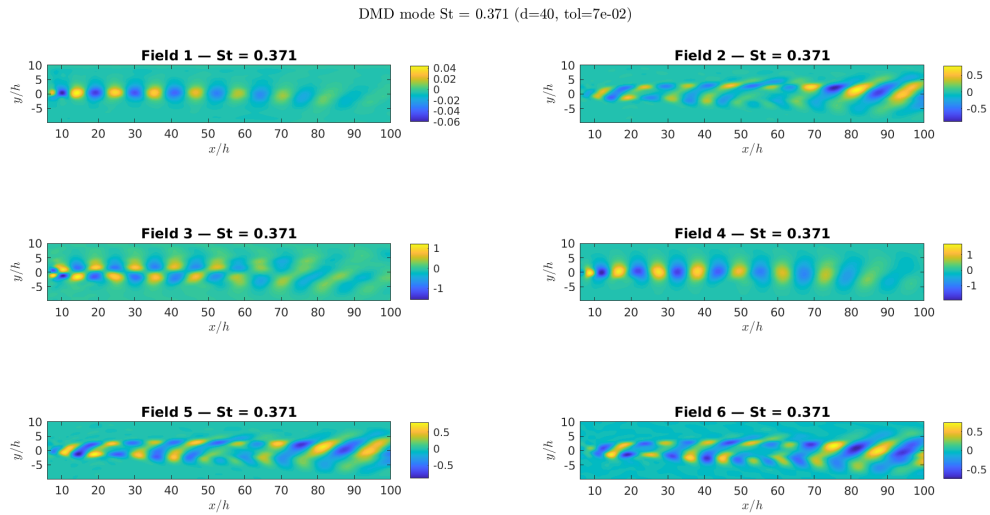


Figure 8: Spatial DMD mode at $St = 0.371$ ($d = 40$, $\varepsilon_{\text{tol}} = 7 \times 10^{-2}$, 1500 K case).

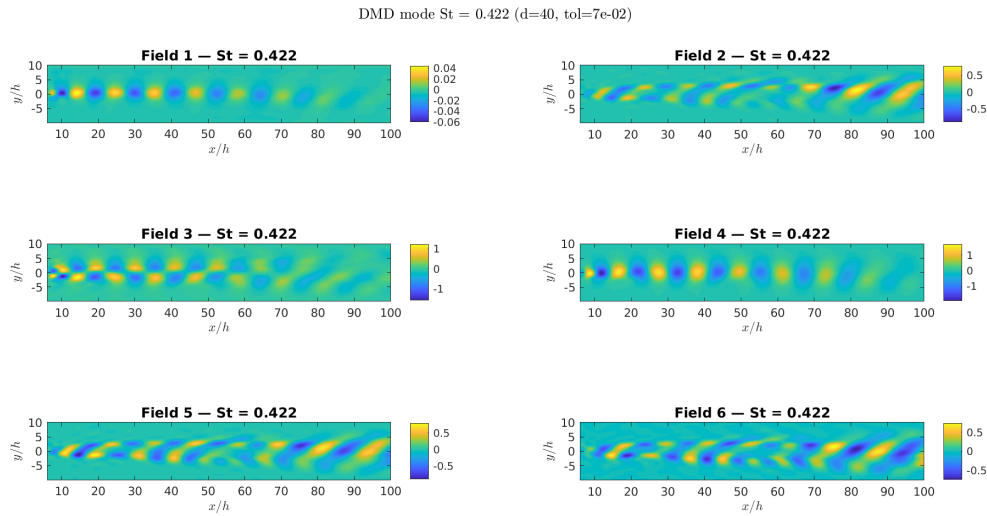


Figure 9: Spatial DMD mode at $St = 0.422$ ($d = 40$, $\varepsilon_{\text{tol}} = 7 \times 10^{-2}$, 1500 K case). This is the highest-amplitude mode in the spectrum.

The mode at $St = 0.422$ (Figure ??) is the dominant mode in terms of amplitude. Its spatial structure closely resembles that of the $St = 0.371$ mode but with a higher spatial frequency. The simultaneous presence of both modes with comparable amplitudes suggests a coupled dynamics, possibly arising from nonlinear interactions within the reactive shear layer.

4.1 Discussion

The mdHODMDit algorithm, in its multidimensional HOSVD-based formulation, successfully extracts physically coherent spatio-temporal modes from the high-dimensional DNS tensors of all three temperature cases without requiring any reshaping of the data into matrix form. The independent truncation of each tensor mode by HOSVD enables a more targeted noise filtering compared to a standard SVD applied to a vectorised snapshot matrix: the variable axis is compressed according to the co-variance structure among the 20 reactive scalars, while the spatial axes are compressed independently along x and y , allowing the method to exploit the anisotropy of the flow geometry.

Variation with preheat temperature The dominant Strouhal numbers identified in the 1500 K reference case lie in the range $St \approx 0.21\text{--}0.42$. [PLACEHOLDER — once results for the 1200 K and 1700 K cases are available, describe how the dominant frequencies shift with preheat temperature. Expected behaviour: an increase in bulk velocity with temperature raises the absolute frequency of the Kelvin–Helmholtz instability; the Strouhal number, which normalises by U_b , may remain approximately constant if the instability scales with the local velocity, or may shift if the density ratio across the shear layer changes significantly with preheat.]

Physical interpretation of the dominant modes The spatial modes show a clear progression from large-scale shear-layer structures at low St to finer-scale alternating patterns at higher St , reflecting the multi-scale nature of the reactive flow dynamics. The modes at $St \approx 0.21\text{--}0.25$ are consistent with the primary convective instability of the non-premixed mixing layer, in which vortices form and grow downstream of the splitter plate. The higher-frequency modes at $St \approx 0.37\text{--}0.42$ likely correspond to secondary instabilities or harmonics of the primary roll-up, and their high amplitude in the 1500 K case indicates a vigorous nonlinear energy transfer between the two frequency bands. The fact that the spatial support of all identified modes is concentrated in the region $x/h \lesssim [PLACEHOLDER]$ suggests that the dominant dynamics are governed by the near-field development of the shear layer rather than by the far-field flame structure.

Comparison with matrix-based HODMD [PLACEHOLDER — if results from a standard (SVD-based, matrix) HODMD are available for the same dataset, compare the number of modes retained, the dominant frequencies, and the reconstruction error. The key expected advantages of the tensor formulation are: (i) a lower compression rank for the same reconstruction accuracy due to mode-wise truncation, and (ii) modes whose spatial structure is directly interpretable as a multi-variable field rather than a single vectorised pattern.]

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